

Developing Optimum Interface Design for On-Screen Chinese Proofreading Tasks

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Abstract. This paper includes a review of some related empirical studies concerning display factors that may contribute to the on-screen Chinese proofreading performance. The effects of typeface, font size, number of text lines, text direction, and copy placement on speed, accuracy, and subjective preferences in past reading tasks are discussed. This paper, in particular, introduces the development of a Chinese Proofreading System for proofreading experiments and delineates some research ideas for identifying the optimum interface design settings for on-screen Chinese proofreading tasks. The results of this research work are expected to provide useful design recommendations to assist in determining the display factor settings and text display layout that would improve work performance and satisfaction in Chinese comparison proofreading tasks.

Keywords: Proofreading, Chinese Reading, Chinese Proofreading System, Usability Evaluation, Subjective preference.

1 Introduction

Proofreading is an important task for reducing input and printing errors to ensure accurate information presentation to readers. The two main ways of proofreading are comparison proofreading and noncomparison proofreading. Comparison proofreading is a critical process in electronic book production and it involves careful reading of the dead copy (original copy) with a live copy and marking deviations (e.g. typeface, page and column breaks) from specifications, misspellings, nonstandard grammar, and other errors in language on the live copy. Noncomparison proofreading is usually done when there is no true dead copy or when the dead copy is only referred to if the proofreader sees something puzzling. An overview of the display factors of typeface, font size, number of lines of text, text direction, and copy placement that are likely to affect speed, accuracy, and subjective preferences in proofreading tasks is presented below.

1.1 Typeface

The display factors of typeface and font size have been shown to affect text readability and reading task efficiency on computer screens [1] - [3]. Research studies on Chinese characters with regard to legibility assessment [4], reading performance [5], and character identification task [6] have been reported. Chi et al. [4] found that Hei characters are the most legible, followed by Ming, Kai, and Li characters. In reading a variety of message signs, Lai [5] found that participants achieved higher accuracy for Hei and Ming styles than for Kai style. Given the diverse results concerning the effects of font type on different perceptual tasks, it is of great interest to attempt to determine the best font type for the proofreading process.

1.2 Font Size

In measuring readability on computer screens as a function of character size, Snyder and Maddox [7] examined the effects of dot element size and character size and showed that smaller characters produced faster English reading speed than larger ones. However, in the same study, larger characters were found to produce faster search times in a search task. Larger characters are generally considered more readable than smaller ones, but it seems likely that there may be an optimum size such that changes in either direction (smaller or larger) will reduce reading performance [1, 8]. For reading or proofreading Chinese characters, the effects of font size are still largely unknown, though it was shown that the legibility of Chinese characters was enhanced by increasing font size [9]. Therefore, it is expected that on the basis of previous studies, larger font size should be preferred and result in better Chinese proofreading performance.

1.3 Number of Lines of Text, Line Spacing, and Line Length

Past research examining the number of lines of text in paged view has produced varied results in reading tasks [10]. Ling and van Schaik [11] investigated the effects of text presentation on information search tasks and found that wider line spacing resulted in better accuracy and faster search times. Previous work on line length was conducted for English reading and the results revealed that line spacing and number of text lines are two opposing factors for determining the optimum line length for reading [12] - [14]. However, the optimum line length for reading Chinese is not yet known. There is thus a need to find out the optimum number of lines, line spacing, and line length for presenting large amount of materials in windowed view in Chinese proofreading task.

1.4 Text Direction

Chinese text has been traditionally written and printed vertically from top to bottom of a page, with columns moving from right to left. However, since the early 20th

century this traditional convention has been largely westernized and, in a lot of printed material today, Chinese characters appear horizontally (usually starting in the top left corner). Sun et al. [15] found that horizontal Chinese characters were read twice as fast as vertical ones. Shih and Goonetilleke [16] revealed that for the Chinese population, a one-row word menu in Chinese was more effective than a one-column word menu in terms of search time and accuracy. Also, in a visual search study, Goonetilleke et al. [17] found that Chinese people used predominantly horizontal search patterns whilst searching for a Chinese target character in a background of distracting characters. Thus, the horizontal reading direction is expected to be more preferred and result in better Chinese proofreading performance than the vertical direction. However, it is believed that in a proofreading task, the text direction factor may interact with the copy placement factor.

1.5 Copy Placement

Copy placement refers to the way in which the live and dead copies are arranged. Anderson [18] recommended that the live copy is always placed nearest to proofreader's pen position. For the common left-right copy placement setting, the dead and live copies are under the left and right hands, respectively for the right-handers. An above-below setting can also be used and a common practice is that the dead copy is placed at the top and the live copy at the bottom [19], such that the live copy is also close to the proofreader's hand. For Chinese characters particularly, a study on the effects of copy placement and its interaction with text direction on proofreading is necessary to help designers create better interfaces for proofreading tasks on computer screens.

2 Methodology

2.1 Experimental Design

This study will examine the ways of improving proofreading performance and proofreader subjective preferences in two phases of research work. In view of the possible influence of the above mentioned display factors on proofreading Chinese, the effects of typeface, font size, text direction, and copy placement on speed, accuracy, and subjective preferences in proofreading will be examined in the first phase of this research. While the first phase is aimed to finding the optimum values or settings for display factors to present printed material on a page, the second phase will be directed at investigating how a large amount of text can be displayed and read effectively using scrolling in the window view method when the material cannot be accommodated within a single viewing page. The proofreading time and performance (number of hits, misses, false alarms and other errors.) will be measured and analyzed. To evaluate subjective preferences, a 10-point Likert scale is used to obtain participants' opinions on attributes of proofreading comfort, ease and fatigue [3].

2.2 Participants

Native Chinese university students with more than five years of computer experience will be invited to participate in each experiment. They will all be right handed in operation of a computer mouse and for handwriting in daily life. Participants will be tested to ensure normal or corrected-to-normal vision with an Orthorator.

2.3 Materials

A number of unrelated Chinese prose passages will be extracted from past examination papers of the Chinese Language subject of the Hong Kong Certificate of Education Examination. Mean passage length, including punctuation marks, will be around 550 characters. An average of five errors will be embedded per passage of a live copy. The errors will include extra word, missing word, wrong word, wrong order, and extra spacing, etc. Both dead and live copies will be presented in a positive polarity condition with black text on a white background.

2.4 Apparatus and Software

A personal computer (Intel Xeon E5506 2.13 GHz) with a 24-inch liquid crystal display monitor and an application program prepared with Microsoft Visual Studio 2008 Professional will be used for stimulus presentation and response capture. At this stage of study, a Visual Basic program (2008 Express Edition) was used to develop the interface of Chinese Proofreading System. An experimenter can adjust the typeface, font size, line length, line spacing, text direction and copy placement according to specific test conditions. The system operation flow is shown in Fig. 1. An illustration of the proofreading interface is shown in Fig. 2. The proofreading system is embedded with five different check functions, which corresponds to the errors of extra word, missing word, wrong word, wrong order and extra spacing. Those check functions were assigned with different colors to stand out their representing errors. The dead copy and live copy are shown on the screen at the same time. The page number of the passage is shown on the top right hand corner to tell the progress of the test. The Next and Undo buttons are for proceeding to next passage and returning to last step respectively. Proofreading time and hitting accuracy for each passage are calculated by default formulas and those data will be stored for analysis. For detail operation of the proofreading system, readers can refer to the section 2.5. Generally, the Chinese Proofreading System contains three main components: User Maintenance, Participant Maintenance and Passage Maintenance. The User Maintenance allows the experimenter to assign experimental conditions to the participants. The Participant Maintenance allows participants to input their personal information. Their test performance is recorded. The Passage Maintenance allows experimenter to add or edit the passages.

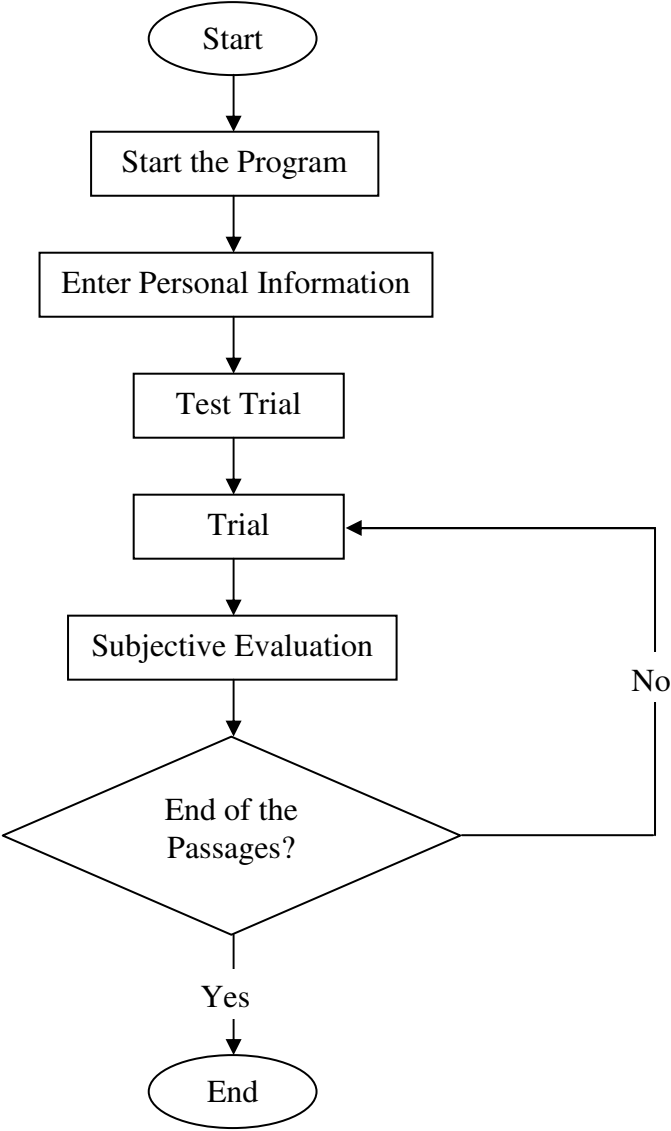


Fig. 1. System Operation Flow

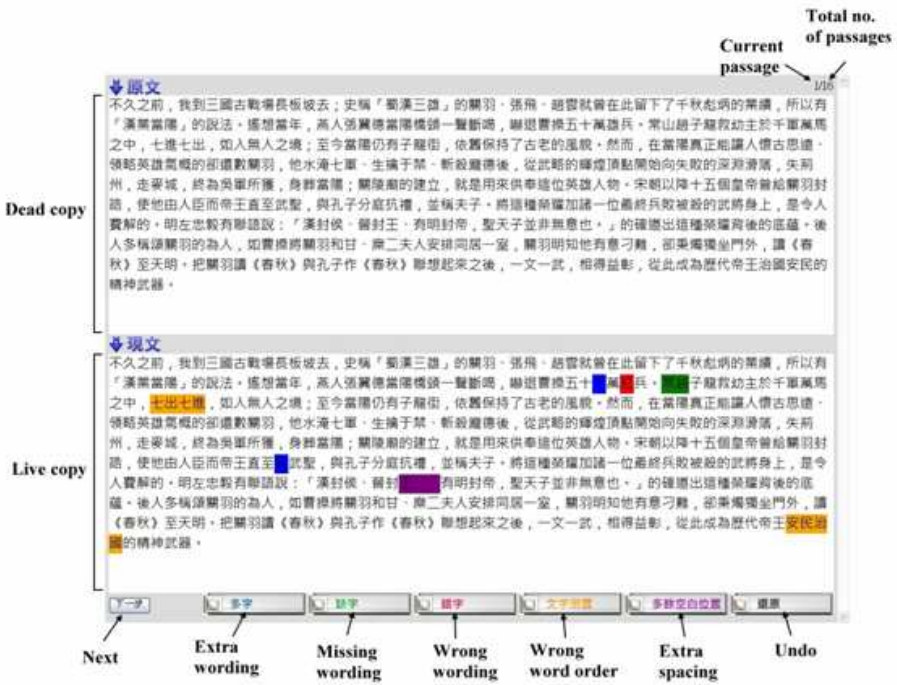


Fig. 2. An illustration of a developed proofreading interface

2.5 Chinese Proofreading System

The purpose of this section is to show a proposed Chinese Proofreading System which will be used for the planned proofreading experiments. During proofreading, participants are allowed to scroll or navigate the two documents independently or jointly together by sliding the scroll wheel in the corresponding document frame areas. An illustration of the proofreading interface is shown in Figure 2. Participants will be asked to compare the live copy against the dead copy and to highlight the errors in the live copy as fast and accurately as they can by using the computer mouse to click the corresponding error type button shown at the bottom of the screen. Then the participants will be asked to evaluate subjective preferences for that condition using a 10-point Likert scale on attributes like comfort (1 = least comfortable, 10 = most comfortable), ease (1 = least easy, 10 = most easy) and fatigue (1 = least fatiguing, 10 = most fatiguing). This process will be repeated until the testing of all experimental conditions is finished. To avoid mental or visual fatigue, a two-minute rest will be given after a certain number of passages are proofread. Participants will take around one to two hours to complete the experiment.

3 Summary

In view of the likely influence of the display factors discussed above on Chinese proofreading, the effects of typeface, font size, text direction, and copy placement on

speed, accuracy, and subjective preferences in proofreading materials displayed in a single page will be examined in the first experiment of this study. A natural extension of this work is to investigate how a large amount of text can be effectively displayed on screen so as to maximize performance and satisfaction of proofreaders when it is not possible to accommodate and present the material within a page of view. In each stage, the design details of an experiment will be carefully considered and confirmed. The experiments will then be conducted to collect participant task times, accuracy, and preference ratings for the different test conditions. The experimental data will be analyzed and the results will be useful for formulating design recommendations and determining the display factor and screen design settings that will improve comparison proofreading performance for Chinese.

Acknowledgments. The work described in this paper was fully supported by a grant from the Research Grants Council of the Hong Kong Special Administrative Region, China (Project No. CityU 110410). The authors thank for the technical support provided by SM Wong.

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